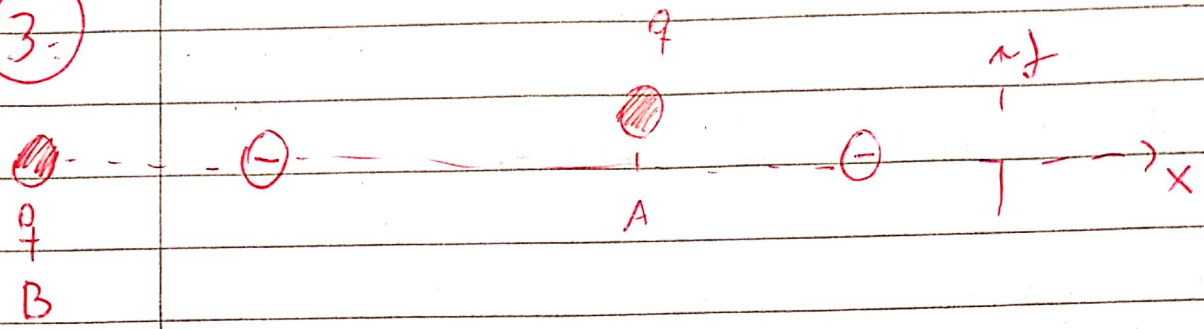
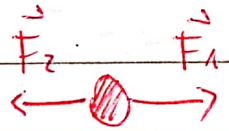
+1

3.




a) A:




$$\vec{F}_2 = 9 \cdot 10^9 \frac{(6{,}25 \cdot 10^{-9})(5 \cdot 10^{-9})}{(0{,}15)^3}(-0{,}15\,\hat{\imath})$$

$$= 1{,}25 \cdot 10^{-5}\,(-\hat{\imath})\ [N]$$

$$\vec{F}_1 = 9 \cdot 10^9 \frac{(12{,}5 \cdot 10^{-9})(5 \cdot 10^{-9})}{(0{,}1)^3}(0{,}1\,\hat{\imath})$$

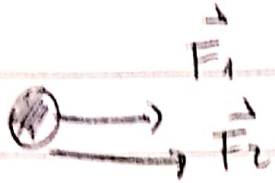
$$= 5{,}625 \cdot 10^{-5}\,(\hat{\imath})\ [N]$$

$$\vec{F}_T = \vec{F}_1 + \vec{F}_2$$

$$= 4{,}375 \cdot 10^{-5}\,(\hat{\imath})\ [N]$$

Escaneado con CamScanner

B)




$$\vec{F_1} = 9 \cdot 10^9 \frac{(12{,}5 \cdot 10^{-9})(5 \cdot 10^{-9})}{(0{,}35)^3} 0{,}35\,\hat{\imath}$$

$$= 4{,}592 \cdot 10^{-6}\ \hat{\imath}\ [N]$$

$$\vec{F_2} = 9 \cdot 10^9 \frac{(6{,}25 \cdot 10^{-9})(5 \cdot 10^{-9})}{(0{,}1)^3} (0{,}1\,\hat{\imath})$$

$$= 2{,}813 \cdot 10^{-5}\ \hat{\imath}\ [N].$$

$$\vec{F_T} = \vec{F_1} + \vec{F_2} \quad ; \quad \vec{F_T} = 3{,}272 \cdot 10^{-5}\ \hat{\imath}\ [N].$$

Escaneado con CamScanner